

SAVOONGA, AK, USA

WMO: 702035

Lat: 63.687N

Lon: 170.493W

Elev: 55

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.1	-27.5	1.2	-16.8	-26.2	1.3	-15.0	42.6	28.1	36.7	23.6	4.5	200	0.991

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	57.1	50.7	54.6	49.6	53.1	48.8	52.4	55.9	50.5	53.8	49.1	52.3	12.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.9	53.3	54.1	48.0	49.5	52.3	46.2	46.3	50.1	21.6	55.9	20.4	53.8	19.7	51.9	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
37.1	32.8	28.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-23.4	56.6	6.1	5.3	-27.8	60.4	-31.4	63.5	-34.8	66.4	-39.3	70.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	5.8	6.0	8.8	17.0	30.0	40.4	47.1	46.7	41.4	33.9	26.4	14.4	(6)
	DBStd	17.73	13.64	15.14	12.57	10.59	6.14	4.90	4.16	3.94	4.82	5.19	7.45	13.47	(7)
	HDD50	8577	1372	1233	1278	991	620	292	109	114	258	498	707	1105	(8)
	HDD65	14016	1837	1653	1743	1441	1085	739	554	567	707	963	1157	1570	(9)
	CDD50	36	0	0	0	0	0	3	20	12	1	0	0	0	(10)
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	12.6	12.4	13.1	11.5	10.7	9.6	9.5	10.2	11.2	12.6	15.5	18.5	16.9	(14)
(15) Precipitation	PrecAvg	22.9	1.7	1.5	1.1	1.0	1.2	1.5	2.4	3.2	2.8	2.4	2.3	1.8	(15)
	PrecMax	34.9	8.1	6.3	3.1	4.0	2.8	3.3	4.2	5.6	4.8	6.0	4.4	4.1	(16)
	PrecMin	13.7	0.0	0.2	0.0	0.2	0.4	0.3	0.7	0.6	1.1	0.5	0.4	0.0	(17)
	PrecStd	4.8	1.6	1.3	0.9	0.7	0.7	0.6	0.8	1.0	1.0	1.2	1.3	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	35.8	35.9	36.5	45.8	63.2	64.2	58.6	54.0	48.4	41.0	36.4	(19)
		MCWB	33.0	32.5	33.3	33.5	41.4	53.3	54.2	52.2	50.4	46.1	35.5	33.4	(20)
	2%	DB	33.7	33.8	33.8	34.5	39.4	54.6	58.7	55.1	51.9	45.9	37.4	34.5	(21)
		MCWB	31.7	31.8	31.7	32.0	35.9	48.0	52.0	49.8	48.2	43.7	34.8	32.5	(22)
	5%	DB	30.3	31.8	31.3	33.9	37.3	50.4	55.1	54.1	49.9	43.4	36.6	33.4	(23)
		MCWB	29.3	30.7	29.9	31.6	34.2	45.5	50.2	49.4	46.7	40.5	34.3	31.7	(24)
	10%	DB	27.3	29.6	28.2	31.8	36.4	47.0	53.8	52.3	48.2	41.0	35.8	30.1	(25)
		MCWB	25.7	28.6	26.7	30.1	33.7	43.3	49.4	48.4	45.7	38.5	33.7	29.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	33.2	33.5	34.1	41.4	53.7	56.3	54.2	50.6	46.5	37.4	34.2	(27)
		MCDB	34.7	34.4	34.6	36.0	45.8	62.8	62.5	57.1	52.8	47.7	38.5	35.3	(28)
	2%	WB	31.6	32.0	31.7	32.4	37.0	48.4	53.1	51.6	48.4	44.3	36.1	32.6	(29)
		MCDB	33.1	33.3	33.2	34.1	39.7	54.0	57.2	54.0	50.5	45.9	37.7	34.1	(30)
	5%	WB	29.2	30.6	29.9	31.6	35.1	45.8	50.9	50.3	47.0	41.4	34.6	31.4	(31)
		MCDB	30.3	31.8	31.1	33.3	37.4	49.7	54.6	52.9	48.9	43.8	36.5	32.9	(32)
	10%	WB	25.7	28.2	27.1	29.8	33.6	43.7	49.5	49.0	46.1	38.8	33.2	29.0	(33)
		MCDB	27.3	29.1	28.2	31.2	35.9	46.8	53.1	51.7	47.8	41.1	35.2	29.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	11.0	9.3	5.7	8.4	8.1	6.7	5.7	5.1	5.9	7.9	(35)
		MCDBR	11.9	10.6	10.4	7.6	7.6	15.2	12.9	9.4	7.6	6.5	6.1	8.3	(36)
	5% WB	MCWBR	11.2	10.0	9.7	6.5	6.2	10.5	8.8	6.9	5.8	5.8	5.7	7.6	(37)
		MCWBR	11.7	10.4	10.2	7.6	7.7	13.9	12.2	8.3	6.8	6.5	5.8	8.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.186	0.270	0.280	0.342	0.375	0.374	0.379	0.399	0.338	0.307	0.408	N/A	(40)
		taud	1.956	2.065	2.154	2.020	2.169	2.384	2.424	2.297	2.441	2.269	2.689	N/A	(41)
	Ebn at Noon		135	202	257	260	263	268	263	243	239	189	100	N/A	(42)
		Edh at Noon	9	21	29	42	40	33	31	32	23	17	9	N/A	(43)
(44) All-Sky Solar Radiation	RadAvg		56	271	803	1301	1409	1688	1346	982	635	265	61	20	(44)
	RadStd		7	40	79	147	143	119	118	103	52	32	7	2	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page